Lecture 7

In this lecture we begin with some fragments of logic (for more details read appendix B of your course notes and try exercises therein) and then begin some study of invertible matrices.

Some logic

Definition 20. *A (mathematical) statement is a sentence which is either true or false but not both.*

Examples $2+2=4$ is a true statement while $2023<0$ is a false statement. Let $z \in \mathbb{C}$, then the expression $z^2+1=0$ is true or false depending on the value of z, similarly let $A, B \in M_{n\times n}(\mathbb{R})$, the expression $AB = BA$ is true or false depending on which matrices A, B are used. We call such expressions such open sentences. Open sentences can be turned into statements by using quantifiers which we shall see shortly in the notes.

Logical connectives

Given (mathematical) statements, we can build more statements by using logical connectives: Conjuction, and ($\wedge$); Disjuction, or ($\vee$); Implication, if ... then ($\implies$), Biconditional, if and only if ($\iff$), Negation, not ($\neg$). Since statements are either true or false, but not both we shall use '1' to represent 'true' and 0 to represent 'false'. Below are the truth tables for our logical connectives.

P	Q	P$\wedge$ Q
0	0	0
0	1	0
1	0	0
1	1	1

P	Q	P $\vee$ Q
0	0	0
0	1	1
1	0	1
1	1	1

P	Q	P $\implies$ Q
0	0	1
0	1	1
1	0	0
1	1	1

P	Q	P $\iff$ Q
0	0	1
0	1	0
1	0	0
1	1	1

P	$\neg$P
0	1
1	0

<u>Converse statement</u>: Given statements P and Q, the converse of statement $P \implies Q$ is the statement $Q \implies P$. *Statement $P \implies Q$ is not logically equivalent to its converse $Q \implies P$.*

Examples The converse of statement '*if* $x=-1$ *then* $x^2=1$' is statement '*if* $x^2=1$ *then* $x=-1$', notice that the given statement is true while its converse is false. Similarly, the converse of statement '*if matrix A is a diagonal matrix then A is an upper triangular matrix*' is the statement '*if A is an upper triangular matrix then A is a diagonal matrix*'. This is to emphasize that the converse of a true statement need not be true, but both may be true for example '*if* $|x|=0$ *then* $x=0$' and its converse '*if* $x=0$ *then* $|x|=0$' are true.

<u>Contrapositive statement</u>: Given statements P and Q, the constrapositive of statement $P \implies Q$ is the statement $\neg Q \implies \neg P$. *Statement $P \implies Q$ is logically equivalent to its converse $\neg Q \implies \neg P$.* Below we show the logical equivalence between the two.

P	Q	$\neg$ P	$\neg$ Q	P $\implies$ Q	$\neg$ Q $\implies$ $\neg$ P	(P $\implies$ Q) $\iff$ ($\neg$ Q $\implies$ $\neg$ P)
0	0	1	1	1	1	1
0	1	1	0	1	1	1
1	0	0	1	0	0	1
1	1	0	0	1	1	1

Example Let $f : \mathbb{R} \to \mathbb{R}$ be a function given by $f(x) = 4x+1$, If we want to show that $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$, we may equivalently show that $f(x_1) = f(x_2) \implies x_1 = x_2$ (this should be very easy to do).

Exercise: Given statements P, Q and R, use a truth table to show that $P \implies (Q \vee R)$ is logically equivalent to $(P \wedge \neg Q) \implies R$.

You will find the above useful for proving statements of type $P \implies (Q \vee R)$, for example if you want to prove that for complex numbers z and w, $zw = 0 \implies z = 0 \vee w = 0$, you may suppose that $zw = 0$ and $z \neq 0$, and then show that $w = 0$. Think or find many more examples of this type across your mathematical background and prove them.

Quantifiers We can turn open sentences into statements by using quantifiers. We have two quantifiers, that is, the universal quantifier 'for all' ($\forall$) and existential quantifier 'there exists' ($\exists$). Given the open sentences above, and using quantifiers we get two statements '*for all complex numbers z, we have* $z^2 + 1 = 0$' and '*there exists a complex number z, such that* $z^2 + 1 = 0$'. The first statement is false, while the second statement is true. Similarly, we have two statements '*for all matrices A and B, we have* $AB = BA$' (which is false) and '*there exists matrices A and B, such that* $AB = BA$' (which is true).

Invertible matrices

Recall from your first-year that an $n \times n$ identity matrix I_n is the diagonal matrix with all the entries in the diagonal equal to 1, that is $a_{ii} = 1$ for all $1 \leq i \leq n$ and $a_{ij} = 0$ for $i \neq j$ with $1 \leq i, j \leq n$. Given any $n \times n$ matrix A, the following identity laws hold:

- $AI_n = A = I_nA$

Using matrix size arguments, one can show that only square matrices can be invertible.

Definition 21. *An $n \times n$ matrix A is invertible if and only if there exists an $n \times n$ matrix B such that $AB = I_n = BA$*

Proposition 7. *If an inverse of an $n \times n$ matrix A exists, then it is unique.*

Proof. Suppose B and B' are inverses of matrix A, then we have:

$$B = BI_n = B(AB') = (BA)B' = I_nB' = B'$$

□

Since the inverse of A is unique when it exists, we shall denote it by A^{-1}.

Proposition 8. *Given two invertible $n \times n$ matrices A and B, then their product AB is also invertible and $(AB)^{-1} = B^{-1}A^{-1}$.*

Proof. We verify this by checking the following (try to also do it independently):

$$(AB)(AB)^{-1} = (AB)(B^{-1}A^{-1}) = A(B(B^{-1}A^{-1})) = A((BB^{-1})A^{-1})) = A(I_nA^{-1}) = AA^{-1} = I_n$$

$$(AB)^{-1}(AB) = (B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}(AB)) = B^{-1}((A^{-1}A)B) = B^{-1}(I_nB^{-1}) = BB^{-1} = I_n$$

□